Sahaj Bhadja

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Summary

Performance infrastructure engineer with 2+ years building **Python**, **C++**, data pipelines, distributed job systems, automated testing, and cloud-native platforms for reliable, measurable software delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Structures, Computer Networks, Cloud Computing, Statistical Methods

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Built data integration pipelines using FastAPI, SQL Server, and cloud storage, processing 5K+ monthly jobs with incremental loads, reconciliation checks, error handling, and CI/CD-backed releases at 99%+ reliability.
- Restructured fault-tolerant asynchronous orchestration using Python workers, explicit job-state transitions, and failure isolation, processing 1K+ jobs while increasing throughput threefold under concurrent production workloads with reliable recovery.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Designed concurrent batch processing and secure file ingestion through AWS RDS and REST APIs, reliably processing 1K+ files per batch with fault-tolerant execution under peak workloads at scale.
- Implemented an AWS-hosted backend serving 150+ instructors and 10K+ daily requests, using CI/CD-driven workflows, production monitoring, and usage tracking to reduce feedback turnaround time by 60% under load.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles*

- Built an end-to-end Python evaluation pipeline processing 100K+ autonomous-driving trajectory logs, improving long-horizon planning by 18% through repeatable experiments, performance analysis, and closed-loop simulation testing at scale.
- Architected a spatial-temporal encoder integrating HD-map geometry and agent histories through multi-head attention, improving closed-loop planning stability by 15% while validating behavior across simulation scenarios at scale.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Engineered cross-platform microservices with automated VM builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS environments with repeatable nightly reporting.
- Optimized API and ETL workflows using OOP patterns, AWS Glue, S3, and SQL, improving response time by 30%, pipeline efficiency by 68%, and storage costs by 15%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Developed a distributed resource-management platform with WebSockets, REST APIs, and eight microservices supporting authentication, billing, notifications, and real-time synchronization of 2M+ records across distributed locations in production.
- Automated Docker-based builds, testing, and staged deployments using Jenkins CI/CD, reducing deployment time by 50% while maintaining modular services and repeatable production-release workflows with quality gates at scale.

Skills

Languages: Python, C++, SQL, Java, JavaScript, TypeScript, Bash
Systems & Performance: Linux, multithreading, distributed systems, operating systems, networking, job orchestration, benchmarking, statistical analysis, regression detection
Data & Observability: ETL, data ingestion, transformation, PostgreSQL, SQL Server, SQLite, query optimization, monitoring, performance metrics, reconciliation
Cloud & Infrastructure: GCP, AWS, Azure, Kubernetes, Docker, Jenkins, GitHub Actions, CI/CD, Git
Engineering: unit testing, integration testing, debugging, profiling, capacity planning, failure isolation, code review, Agile/Scrum, technical documentation

Projects

BenchFleet | Python, SQLite, benchmark orchestration, regression detection

- Built and tested capacity-aware benchmark infrastructure with concurrent job scheduling, latency and memory capture, SQLite metric ingestion, Linux host parsing, p95 summaries, and effect-size regression gates at scale.

CPU-scheduling Simulator | Live | C++, React, JavaScript

- Implemented six CPU scheduling algorithms with dynamic execution metrics and Gantt visualization across 1K+ process combinations, reducing average turnaround time by 25% while testing systems-level resource allocation.

FinNexus | Live | TypeScript, Kubernetes, MongoDB, WebSockets

- Built a Kubernetes-deployed platform supporting 50K+ users and 1M+ transactions with real-time synchronization, advanced filtering, secure APIs, and automated CI/CD across reliable production workflows at production scale.